

# POWER BI DAX REPORT- PRODUCTIVITY MATRIX

## PROJECT OVERVIEW:

The Power BI DAX Productivity & Screen Time Dashboard is designed to analyse and visualize user behaviour across productivity, study hours, screen time, and sleep quality. Using advanced DAX measures and calculated columns, the dashboard consolidates large-scale user activity data into actionable insights.

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## OBJECTIVE:

- ✚ Track average productivity across all users.
  - ✚ Aggregate total productivity for all users.
  - ✚ Measure total hours of screen usage.
  - ✚ Monitor average free time per user.
  - ✚ Calculate average active engagement time.
  - ✚ Identify number of high productivity users.
  - ✚ Identify number of low productivity users.
  - ✚ Track users with healthy sleep patterns.
  - ✚ Measure productivity of high screen time users.
  - ✚ Measure productivity of low screen time users.
  - ✚ Measure productivity of good sleep users.
  - ✚ Track study hours of top 5 users.
  - ✚ Track screen time of top 5 users.
  - ✚ Measure total screen time of high addiction users.
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## DATA SOURCES:

- ✚ User activity logs (productivity, screen time, study hours, sleep records)
  - ✚ Aggregated dataset with 10,000+ records
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## KEY METRICS (KPIs):

- Average Productivity: 37.61
  - Total Productivity: 225.68K
  - Total Screen Time: 41.52K
  - Average Free Time: 6.50 hrs
  - Average Active Time: 10.99 hrs
  - High Productivity Users: 1,376
  - Low Productivity Users: 3,445
  - Users with Good Sleep: 1,259
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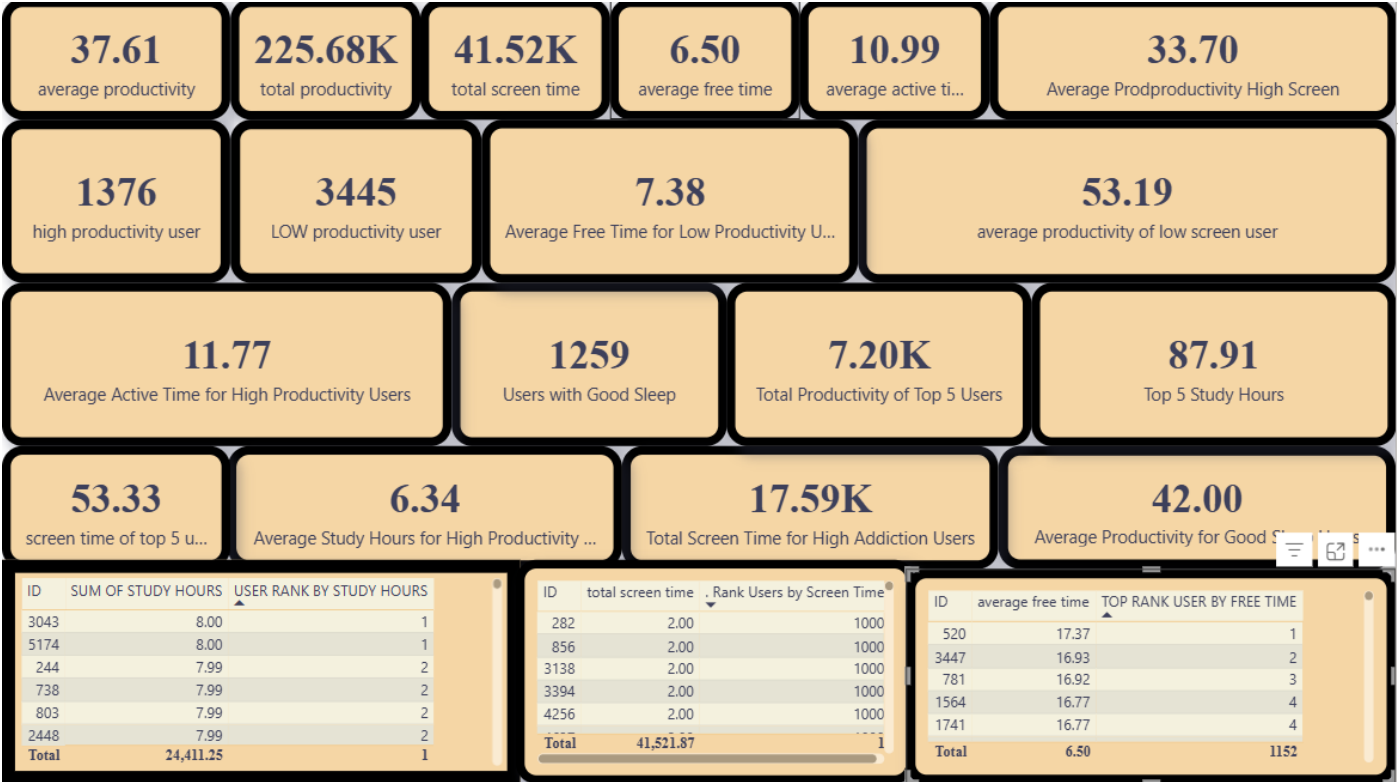
## DAX FORMULA:

Here's a **clean list of DAX formulas** used for your dashboard, written point-by-point:

- Average Productivity = AVERAGE('UserData'[Productivity])
  - Total Productivity = SUM('UserData'[Productivity])
  - High Productivity Users = CALCULATE(COUNTROWS('UserData'), FILTER('UserData','UserData'[Productivity] >= 50))
  - Low Productivity Users = CALCULATE(COUNTROWS('UserData'), FILTER('UserData','UserData'[Productivity] < 50))
  - Total Screen Time = SUM('UserData'[ScreenTime])
  - Average Screen Time = AVERAGE('UserData'[ScreenTime])
  - High Screen Productivity = CALCULATE(AVERAGE('UserData'[Productivity]), FILTER('UserData','UserData'[ScreenTime] > 8))
  - Low Screen Productivity = CALCULATE(AVERAGE('UserData'[Productivity]), FILTER('UserData','UserData'[ScreenTime] <= 8))
  - High Addiction Screen Time = CALCULATE(SUM('UserData'[ScreenTime]), FILTER('UserData','UserData'[ScreenTime] > 12))
  - Total Study Hours = SUM('UserData'[StudyHours])
  - Average Study Hours = AVERAGE('UserData'[StudyHours])
  - Top 5 Study Hours = TOPN (5, SUMMARIZE('UserData','UserData'[UserID],"StudyHours", SUM('UserData'[StudyHours])), [StudyHours], DESC)
  - Good Sleep Users = CALCULATE(COUNTROWS('UserData'), FILTER('UserData','UserData'[SleepQuality] = "Good"))
  - Productivity Good Sleep = CALCULATE(AVERAGE('UserData'[Productivity]), FILTER('UserData','UserData'[SleepQuality] = "Good"))
  - Average Free Time = AVERAGE('UserData'[FreeTime])
  - Average Active Time = AVERAGE ('UserData'[Active Time])
  - Top Free Time Users = TOPN (5, SUMMARIZE('UserData','UserData'[UserID],"FreeTime", AVERAGE('UserData'[FreeTime])), [FreeTime], DESC)
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VISUALS USED:

- KPI Cards (Productivity, Screen Time, Sleep, Study Hours)
- Bar Charts (High vs Low Productivity, Screen Time Distribution)
- Tables (User Rankings)
- Comparative Metrics (Good Sleep vs Poor Sleep Users)



CONCLUSION:

The DAX-powered dashboard successfully consolidates complex user behaviour data into clear, actionable insights. Analysis shows that **low screen time and good sleep patterns strongly correlate with higher productivity**, while excessive screen usage reduces performance. Study hours emerge as a key driver of discipline, with top users setting benchmarks for consistency. By leveraging DAX measures for segmentation, ranking, and KPI tracking, the dashboard provides organizations with a reliable tool to **monitor productivity trends, identify risk groups, and promote healthier digital habits**.